

Chapter 1

Foundations of Information, Data and Computer Security

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1 Introduction to Information Security

Modern organizations depend heavily on digital information to perform their daily activities. Information systems support communication, financial operations, healthcare services, industrial processes, education, and government activities. As the amount and importance of digital information increases, protecting this information against unauthorized access, modification, destruction, or interruption has become a fundamental requirement.

Information security is the discipline concerned with protecting information and information systems from threats while ensuring that authorized users can access resources when required. It combines technical measures, organizational policies, and human practices to preserve the trustworthiness and proper functioning of computer systems.

Definition 1. *Information Security* is the practice of protecting information and related systems from unauthorized access, use, disclosure, disruption, modification, or destruction while maintaining confidentiality, integrity, and availability of information resources [1, 2, 3].

The development of information security has followed the evolution of computing technology. Early computer security focused mainly on protecting individual computer systems. With the growth of networks, distributed systems, cloud computing, and interconnected services, security expanded to include communication protection, identity management, data protection, and organizational risk management [2].

1.1 Computer Security, Network Security and Information Security

Computer security focuses on protecting computing systems, including hardware, software, operating systems, and stored information. It provides measures designed to prevent unauthorized access and reduce the possibility of system compromise.

Network security extends these principles to communication environments. It protects information while it is transmitted between systems and prevents attacks against network infrastructure and services.

Information security represents the broader concept. It includes the protection of all forms of information, whether stored electronically, transmitted through networks, or maintained through organizational processes [4].

Concept	Description
Computer Security	Protection of computer systems and their resources against unauthorized access and malicious activities.
Network Security	Protection of data communication and network infrastructure from attacks and misuse.
Information Security	Protection of information assets through technical, administrative, and physical security measures.

Table 1: Relationship between computer security, network security, and information security.

Information security is therefore not limited to technology. A secure environment requires cooperation between security tools, procedures, policies, and users.

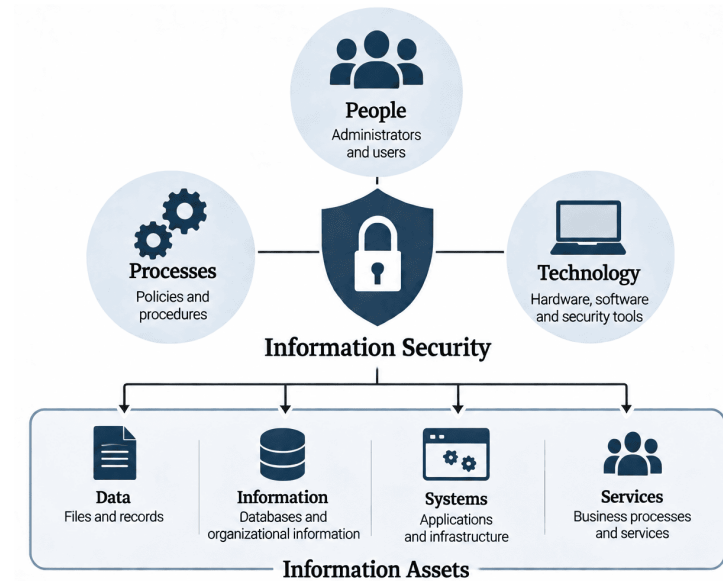


Figure 1: Information security combines technology, processes, and people to protect valuable information assets.

2 Information, Data, and Knowledge

2.1 Data

Data represents raw facts, observations, measurements, or symbols that have not yet been interpreted. Examples include numbers, text, images, sensor readings, and transaction records [5].

A single data element usually has limited meaning. For example, the value “85” does not provide useful information unless its context is known. It may represent a temperature measurement, a student’s grade, or a percentage value.

2.2 Information

Information is data that has been processed, organized, and interpreted to provide meaning. Information supports decision-making and enables organizations to perform their activities effectively [3].

For example, a collection of temperature readings combined with location and time details can become meaningful weather information.

The transformation process can be represented as:

$$\text{Data} \rightarrow \text{Processing} \rightarrow \text{Information}$$

2.3 Knowledge

Knowledge represents the understanding gained from information and experience. Knowledge allows individuals and organizations to interpret information and make appropriate decisions [5].

For example, historical security information can help an organization understand attack patterns and improve future protection strategies.

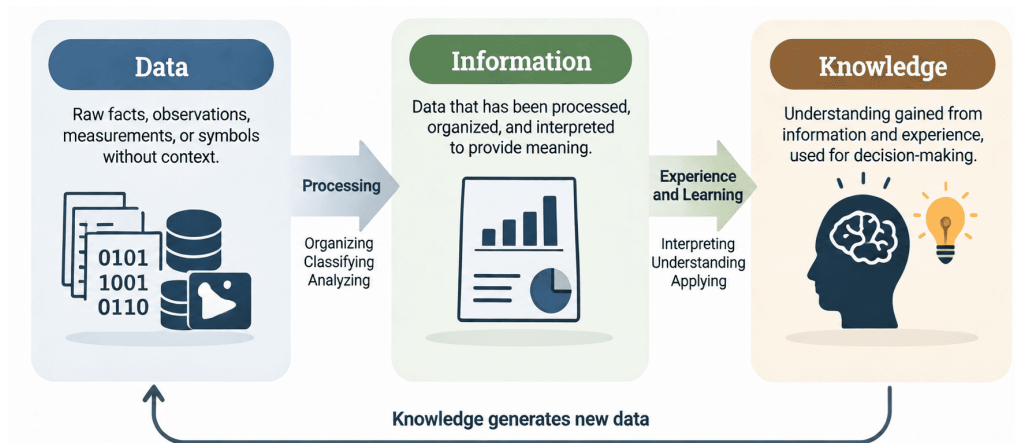


Figure 2: Relationship between data, information, and knowledge

3 Evolution of Computer Security to Cybersecurity

Computer security has evolved significantly as computing technology has changed. Early computer systems were isolated and mainly required protection against physical damage and unauthorized physical access. Modern environments involve global networks, cloud platforms, mobile devices, and Internet-connected systems, requiring a broader security approach [1].

3.1 Historical Development

The evolution of computer security can be divided into several major phases.

3.1.1 Mainframe Security

During the mainframe era, computers were expensive centralized systems used mainly by governments, research institutions, and large organizations. Security focused primarily on physical protection, controlled access to computing facilities, and ensuring system reliability.

3.1.2 Personal Computer Security

The introduction of personal computers expanded access to computing resources. Security concerns shifted toward protecting individual systems from malicious software, unauthorized modification, and data loss.

3.1.3 Network Security

As computers became interconnected through local networks, protecting communication between systems became increasingly important. Organizations began focusing on protecting shared resources and controlling network-based access.

3.1.4 Internet Security

The rapid growth of the Internet created a global environment where systems could communicate worldwide. Security challenges expanded because attackers could target systems remotely across international networks [6].

3.1.5 Cloud and IoT Security

Modern computing relies heavily on cloud services and Internet of Things (IoT) devices. Security now includes protecting distributed systems, connected devices, and large volumes of digital information [7].

3.1.6 Modern Cybersecurity

Cybersecurity represents the current evolution of computer security. It combines technical, organizational, and human approaches to protect digital assets and maintain resilience against constantly changing threats [8].

Table 2: Evolution of Computer Security

Period	Security Focus
1960–1980	Physical access protection and system reliability
1980–1995	Personal computer protection and malware prevention
1995–2010	Network and Internet security
2010–Present	Cloud, mobile, IoT security, and cyber resilience

4 Security Foundations: Assets, Dependability and Requirements

Information security begins with identifying what must be protected. These valuable elements are called assets. Assets may include data, software, hardware, services, users, and organizational knowledge.

Definition 2. *Information Asset* is any resource that has value for an organization and requires protection against threats. An information asset may include databases, documents, applications, computing resources, communication systems, or business processes [1].

4.1 Information Classification

Organizations store information with different sensitivity levels. Information classification helps determine the appropriate level of protection [9].

Common classification categories include:

- **Public Information:** Information that can be freely distributed.
- **Internal Information:** Information intended only for organizational use.
- **Confidential Information:** Sensitive information requiring stronger protection.
- **Restricted Information:** Highly sensitive information requiring the highest protection level.

Table 3: Information Classification Examples

Classification	Example	Protection Level
Public	Published reports	Low
Internal	Organization documents	Medium
Confidential	Customer data	High
Restricted	Credentials and secrets	Very High

An effective security strategy requires understanding the characteristics of dependable systems. Dependability describes the ability of a system to provide correct and reliable service while resisting failures and disruptions [2].

The main dependability properties include:

- **Reliability:** the ability of a system to operate correctly over a specified period.
- **Availability:** the ability of a system or resource to remain accessible when required by authorized users.

- **Maintainability:** the ability to restore or modify a system efficiently.
- **Safety:** the ability to prevent harmful consequences caused by system failures.

Security requirements define what protection must be provided to ensure that information and systems remain trustworthy [1].

Security Requirement	Purpose
Confidentiality	Prevent unauthorized disclosure of information.
Integrity	Prevent unauthorized data modification.
Availability	Ensure authorized access to resources when needed.
Authentication	Verify the identity of users or systems.
Accountability	Ensure actions can be traced to responsible entities.

Table 4: Basic information security requirements.

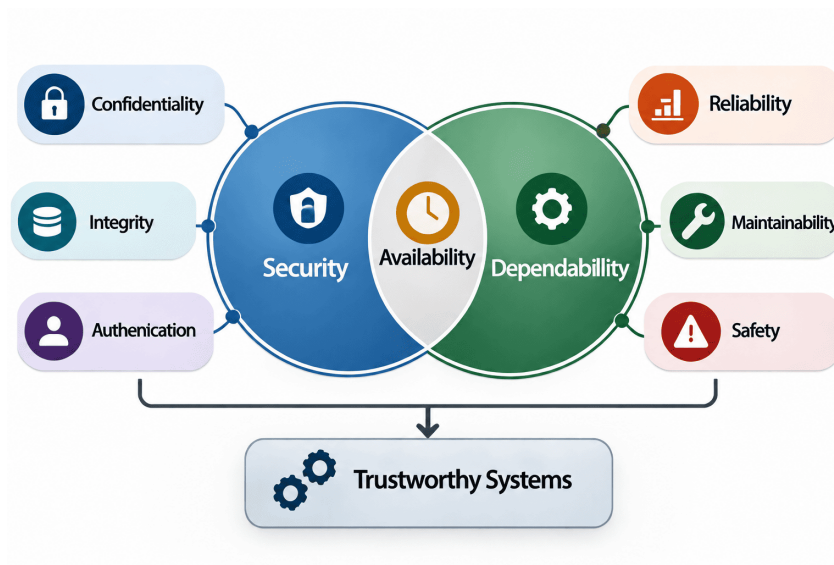


Figure 3: Dependability and security are related concepts that support reliable and trustworthy systems.

5 Security Concepts: Threats, Vulnerabilities, Attacks and Risk

A fundamental objective of information security is understanding how security failures occur. Security incidents usually result from the interaction between valuable assets, potential threats, existing vulnerabilities, and successful attacks. These concepts provide the basic vocabulary required to analyze and manage security problems.

Definition 3. *Threat* is a potential event, action, or circumstance that may cause harm to an information asset, system, or organization. A threat may originate from malicious actors, accidental actions, technical failures, or environmental events [3].

Threats can have different origins. Human-related threats include mistakes, carelessness, malicious insiders, and social engineering attacks. Technical threats include software failures, hardware failures, and vulnerabilities in system design. Environmental threats include natural disasters, power failures, and physical damage [6].

Threat Category	Examples
Human Errors	Incorrect configuration, accidental deletion, weak security practices.
Malicious Activities	Unauthorized access, malware distribution, data theft.
Software and Hardware Failures	Programming errors, system crashes, equipment failures.
Environmental Events	Fire, flooding, electrical problems, natural disasters.

Table 5: Main categories of information security threats.

Definition 4. *Vulnerability* is a weakness or deficiency in hardware, software, configuration, procedures, or human behavior that can be exploited by a threat to compromise the confidentiality, integrity, or availability of information [4].

Examples of vulnerabilities include outdated software, weak authentication practices, incorrect permissions, insecure configurations, and lack of user awareness. The presence of a vulnerability does not automatically create a security incident; exploitation by a threat actor is required.

Definition 5. *Security Attack* is any intentional or unintentional action that attempts to compromise the security of information or systems. An attack may target data, services, applications, users, or infrastructure [1].

A successful attack usually follows a sequence:

1. Identification of a target or valuable asset.
2. Discovery of a weakness or vulnerability.
3. Selection of an appropriate attack method.
4. Exploitation of the weakness.
5. Achievement of an unauthorized objective.

Definition 6. *Risk* represents the possibility that a threat will exploit a vulnerability and produce harmful consequences. Risk evaluation considers both the likelihood of an incident occurring and the impact that the incident may create [8].

A simplified representation of risk is:

$$\text{Risk} = \text{Likelihood} \times \text{Impact} \quad (1)$$

Security management does not attempt to eliminate all risks because complete elimination is usually impossible. Instead, organizations identify risks, evaluate their importance, and apply appropriate security controls to reduce them to an acceptable level.

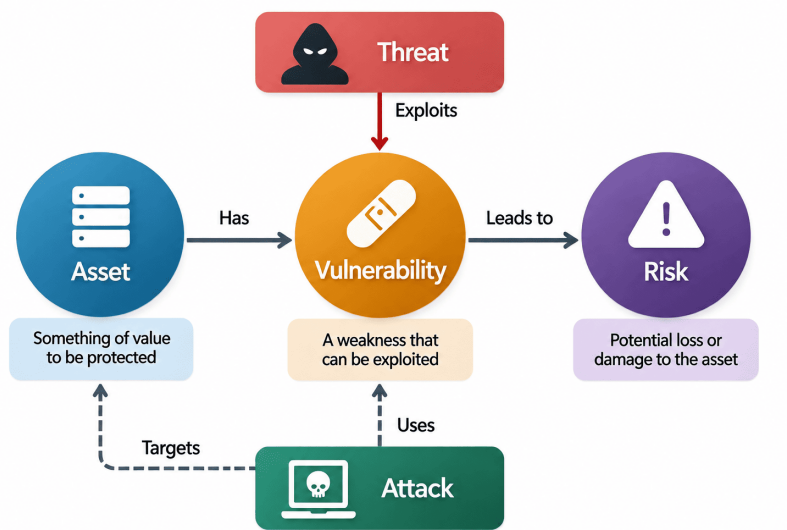


Figure 4: Relationship between assets, threats, vulnerabilities, attacks, and security risk.

6 Threat Actors and Motivations

Threat actors are individuals, groups, or organizations capable of causing harm to information systems [6].

Understanding threat actors helps organizations identify possible risks.

6.1 Cybercriminals

Cybercriminals usually seek financial benefit through activities such as information theft or fraud.

6.2 Hacktivists

Hactivists use cyber activities to promote political or social objectives [7].

6.3 Insiders

Insiders are individuals with legitimate access who may intentionally or accidentally cause security problems.

6.4 Nation-State Actors

Nation-state actors conduct cyber activities for intelligence, strategic, or political objectives.

6.5 Accidental Users

Users may create security problems through mistakes, poor practices, or lack of security awareness [3].

Table 6: Threat Actors and Motivations

Actor	Motivation
Cybercriminal	Financial gain
Insider	Revenge, profit, or mistakes
Nation-state Actor	Intelligence and strategic goals
Hactivist	Political objectives
Accidental User	Human error

7 Security Attacks and Security Objectives

Security attacks represent attempts to violate security objectives. Attacks can be classified according to their effect on information and systems. A common classification divides attacks into passive attacks and active attacks [10].

7.1 Passive Attacks

A passive attack attempts to obtain information without modifying system resources or affecting normal operations. The attacker mainly focuses on observing or collecting information [10].

Examples include:

- Eavesdropping on communications.
- Monitoring network traffic.
- Collecting information about systems and users.

Passive attacks are difficult to detect because they may not change the state of the targeted system. Protection usually depends on preventive measures such as encryption and secure communication techniques.

7.2 Active Attacks

An active attack attempts to modify, disrupt, or damage information systems. Unlike passive attacks, active attacks affect system operation and may create visible changes.

Examples include:

- Unauthorized modification of data.
- Denial of service.
- Malware installation.
- Identity impersonation.

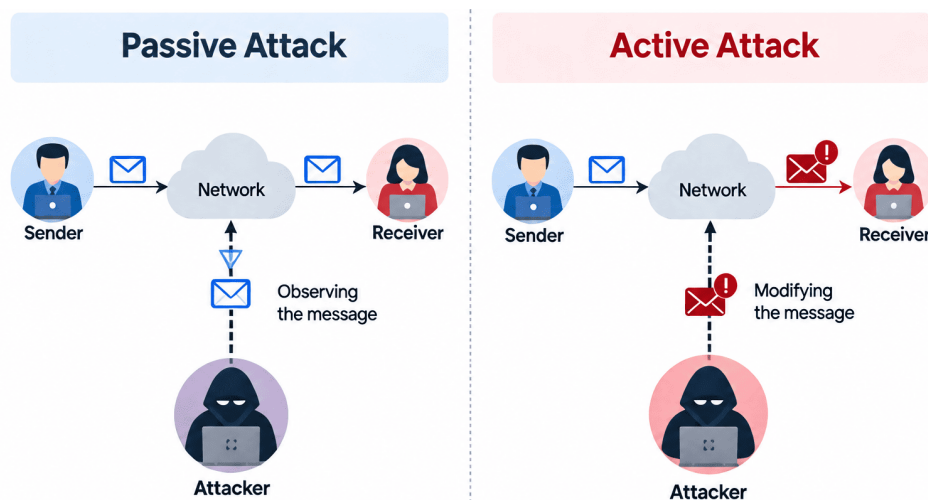


Figure 5: Passive attacks observe information while active attacks modify or disrupt system operation.

7.3 Classical Attack Categories

- **Interception:** Unauthorized access to information, resulting in a loss of confidentiality.
- **Modification:** Unauthorized alteration of information, resulting in a loss of integrity.
- **Interruption:** Disruption of system resources or services, resulting in a loss of availability.
- **Fabrication:** Unauthorized creation or insertion of false information into a system.

7.4 Security Objectives

Information security is commonly based on a set of fundamental objectives that define what protection should achieve [1].

7.4.1 Confidentiality

Confidentiality ensures that information is accessible only to authorized entities. It protects sensitive information from unauthorized disclosure or exposure.

Examples of confidentiality protection include access restrictions, encryption, and secure authentication mechanisms.

7.4.2 Integrity

Integrity ensures that information remains accurate, complete, and protected from unauthorized modification. A system with integrity protection allows users to trust that received information has not been altered improperly.

7.4.3 Availability

Availability ensures that authorized users can access systems, applications, and information whenever required. Availability protection includes backup, redundancy, monitoring, and recovery procedures.



Figure 6: The CIA Triad represents the three fundamental objectives of information security: confidentiality, integrity, and availability.

7.4.4 Authentication, Authorization and Accountability (AAA)

The AAA model provides a structured approach to controlling and tracking access to information systems. Authentication establishes identity, authorization determines permitted actions, and accountability records activities so that actions can be associated with responsible entities.

- **Authentication:** Authentication provides confidence that an entity is who it claims to be. Authentication is required before granting access to protected resources [4].
- **Authorization:** Authorization determines which actions an authenticated entity is permitted to perform. Authentication establishes identity, whereas authorization applies permissions and policies to determine what that identity may access or modify.
- **Accountability:** Accountability ensures that actions performed within an information system can be associated with responsible entities. It supports auditing, investigation, and security analysis by providing evidence about who performed an action and when it occurred.

7.4.5 Non-Repudiation

Non-repudiation prevents an entity from denying a previous action or transaction. It supports accountability and trust in electronic communications [10].

7.4.6 Access Control

Access control determines which users or systems are allowed to access resources and what operations they may perform. Detailed access control models are studied in later chapters.

8 Security Services and Security Controls

Security objectives are achieved through security services and security controls. These concepts describe the methods organizations use to protect information and systems.

Definition 7. *Security Service* is a capability provided by a system to support one or more security objectives. Security services protect resources by reducing the possibility of unauthorized access, misuse, or damage [10].

Definition 8. *Security Control* is a safeguard or countermeasure designed to prevent, detect, respond to, or recover from security threats. Controls may be technical, administrative, or physical [8].

Security controls are commonly grouped into three major categories:

Control Type	Description
Preventive Controls	Designed to stop security incidents before they occur, such as access restrictions and security policies.
Detective Controls	Designed to identify suspicious activity or security incidents, such as monitoring and logging systems.
Corrective Controls	Designed to restore normal operation after an incident, such as backups and recovery procedures.

Table 7: Categories of security controls.

The selection of controls depends on organizational requirements, available resources, and the level of risk that must be managed.

Security Objective	Example Controls
Confidentiality	Encryption, authentication, access restrictions.
Integrity	Hashing, audit records, change management.
Availability	Backup systems, redundancy, disaster recovery.
Authentication	Passwords, multi-factor authentication, identity verification.

Table 8: Examples of controls supporting security objectives.

9 Security Mechanisms

Security mechanisms are safeguards used to implement security objectives and protect information systems against identified threats. Common mechanisms include **encryption**

to protect information from unauthorized disclosure, **digital signatures** to support authenticity and integrity, and **integrity mechanisms** to detect unauthorized modification. Authentication mechanisms verify the identity of users or systems before access is granted. **Monitoring mechanisms** help identify unusual activities and security events, while **recovery mechanisms** restore systems and information following failures or security incidents. Together, these mechanisms provide practical support for maintaining secure and dependable information systems. Detailed cryptographic algorithms and specialized security mechanisms are discussed in later chapters.

10 Types of Information Security

Information security is a broad discipline that includes multiple protection domains. Modern organizations require security measures that cover physical resources, human activities, computer systems, networks, applications, and data. Each security domain addresses specific risks while contributing to the overall protection strategy [6].

Definition 9. Information Security Domain *is a specific area of protection concerned with securing a particular type of resource, environment, or activity within an organization. Security domains work together to provide comprehensive protection against different categories of threats.*

10.1 Physical Security

Physical security protects buildings, equipment, facilities, and hardware resources from unauthorized access, damage, theft, or environmental events [3].

Examples of physical security controls include:

- Security guards and controlled access areas.
- Surveillance cameras.
- Access badges and biometric systems.
- Environmental protection systems.

Physical security is essential because unauthorized physical access can bypass many technical security mechanisms.

10.2 Personnel Security

Personnel security focuses on protecting information systems from risks related to individuals. Since users interact with technology and information resources, human behavior has a significant impact on security [5].

Personnel security includes:

- Security awareness training.
- Employee background verification.
- Clear security responsibilities.
- Procedures for changing or removing user access.

10.3 Host and System Security

Host security protects individual computing systems, including operating systems, servers, and workstations.

Important protection measures include:

- Secure system configuration.
- Software updates and security patches.
- Malware protection.
- System monitoring.

Operating system security is particularly important because all applications and services depend on the underlying system environment.

10.4 Network Security

Network security protects communication infrastructure and data exchanged between systems. It aims to prevent unauthorized access, interception, and network-based attacks [10].

Network security concepts include:

- Secure communication.
- Network segmentation.
- Traffic monitoring.
- Firewall protection.

Detailed network attacks and defense mechanisms are covered in later chapters.

10.5 Application Security

Application security focuses on protecting software applications from design weaknesses, implementation errors, and unauthorized use.

Application security includes:

- Secure software development.
- Input validation.
- Vulnerability management.
- Application testing.

10.6 Data Security

Data security protects information throughout its lifecycle, including creation, storage, transmission, and disposal.

Important data protection principles include:

- Preventing unauthorized disclosure.
- Maintaining data accuracy.
- Ensuring proper access.
- Protecting sensitive information.

10.7 Cloud Security

Cloud security addresses the protection of resources hosted in cloud computing environments. It includes protecting cloud data, services, applications, and user identities.

Cloud security requires understanding shared responsibility between cloud providers and customers.



Figure 7: Information security consists of multiple domains working together to protect organizational assets.

11 Modern Cybersecurity Challenges

The modern digital environment has introduced new security challenges. Organizations now operate complex infrastructures involving cloud services, mobile devices, Internet-connected systems, and large volumes of sensitive data.

Cybersecurity challenges continue to evolve because attackers continuously develop new techniques while organizations increase their dependence on digital technologies.

11.1 Artificial Intelligence and Security

Artificial intelligence (AI) is becoming increasingly relevant to information security because it can support both offensive and defensive activities. Its security implications should therefore be considered from two complementary perspectives: AI-assisted attacks and AI-assisted defense.

- **AI-Assisted Attacks:** The use of artificial intelligence by adversaries to accelerate analysis, automate malicious activities, generate deceptive content, or increase the scale and efficiency of existing attack techniques.
- **AI-Assisted Defense:** The use of artificial intelligence to support security monitoring, identify anomalous activities, analyze security data, and assist professionals in detecting and responding to potential threats.

AI-assisted attacks can accelerate and scale existing techniques, while AI-assisted defense can process security data and identify suspicious patterns. Both require human oversight because automated outputs may contain errors or incomplete information and context.

11.2 Malware

Definition 10. *Malware is malicious software designed to perform unauthorized actions that compromise the confidentiality, integrity, or availability of systems and information. Malware includes different categories such as viruses, worms, trojans, spyware, and ransomware [4].*

Common malware categories include:

Malware Type	Description
Virus	Malicious code that attaches itself to legitimate files or programs and spreads when executed.
Worm	A self-replicating program that spreads through networks without requiring a host file.
Trojan Horse	Malicious software disguised as a legitimate application.
Spyware	Software designed to secretly collect information about users or systems.
Ransomware	Malware that prevents access to data or systems and demands payment for restoration.

Table 9: Examples of malware categories.

11.3 Social Engineering

Definition 11. *Social Engineering is a security threat that uses psychological manipulation to influence individuals into revealing information, performing actions, or bypassing security procedures [5].*

Social engineering demonstrates that humans remain an important component of security. Technical controls alone cannot protect organizations if users are not aware of security risks. Examples include:

- Phishing messages.
- Impersonation attempts.
- Fraudulent requests for information.

11.4 Data Breaches

A data breach occurs when sensitive information is exposed, stolen, or accessed without authorization. Data breaches may affect individuals, organizations, and public institutions [7].

Common causes include:

- Weak authentication.
- Unpatched vulnerabilities.
- Human errors.
- Misconfigured systems.

11.5 Cloud and Internet of Things Security

Cloud computing and Internet of Things (IoT) technologies provide important benefits but introduce additional security challenges [8]. Examples include:

- Large numbers of connected devices.
- Difficulties managing identities.
- Increased attack surfaces.
- Dependence on third-party services.

Security professionals must continuously evaluate new technologies and adapt protection strategies according to emerging risks [7].

12 Chapter Summary and References

12.1 Chapter Summary

This chapter introduced the fundamental concepts required to understand information, data, and computer security. Information security was presented as a discipline that protects valuable resources through technical, administrative, and physical measures.

The chapter explained the relationship between assets, threats, vulnerabilities, attacks, and risks. Understanding these concepts provides the foundation for security analysis and risk management. The main security objectives were introduced:

- Confidentiality protects information from unauthorized disclosure.

- Integrity protects information from unauthorized modification.
- Availability ensures access to resources when required.
- Authentication verifies identities.
- Non-repudiation supports accountability.

The chapter also presented security controls and security domains, showing that effective protection requires cooperation between technology, processes, and people.

Future chapters will examine specialized security topics including access control, cryptography, network security, operating system security, firewalls, and intrusion detection systems.

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